

A water-electricity nexus model to analyze thermoelectricity supply reliability under environmental regulations and economic penalties during drought events

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ABSTRACT

A reliable thermoelectricity supply system requires available cooling water. Recent electricity supply failures that have occurred all over the world during drought events indicate that the thermoelectricity system is vulnerable. However, knowledge and policy gaps remain regarding water-electricity interactions at the watershed scale. In this study, a water-electricity nexus model is developed to facilitate electricity generation decisions and environmental policy designs involving cooling water consumption and temperature controls. This nexus model consists of three modules, a physically based hydrologic module, a decentralized optimization module, and a thermoelectricity generation module, which are closely integrated at the daily time step. The model is used to quantify tradeoffs between environmental impacts and potential electricity supplies at the power plant and watershed levels. A case study is conducted to demonstrate the model application for a watershed-wide, coordinated water-energy policy design under different combinations of environmental regulations and economic penalties during a drought event.

1. Introduction

The reliability and future expansion of the thermoelectricity supply system are vulnerable to inadequate cooling water availability (Tidwell et al., 2012; Zhou et al., 2018). Cooling of thermal power plants requires a significant amount of water withdrawal and consumption during the electricity generation process (Meldrum et al., 2013). An estimated 13 billion m^3 of fresh water is consumed annually for energy production all over the world (Spang et al., 2014). The United States and China account for the most cooling water use. In the US, 45% of water withdrawal, of which approximately 3% is consumed, is used for cooling (Maupin et al., 2010), a percentage that is even higher than water withdrawal used for irrigation. The 1,290 thermoelectric power plants in the US withdraw and consume an estimated 488 and 13 million m^3 of cooling water per day, respectively (Diehl and Harris, 2014). This amount of water can

support hundreds of cities the size of New York City each day (Averyt et al., 2011). In China, 178 and 13 million m^3 of water are withdrawn and consumed per day for thermoelectricity generation, respectively (Liao et al., 2016; Zheng et al., 2016; Zhang et al., 2017), and are projected to exceed 767 and 41 million m^3 by 2050, possibly violating the government quotas on water if no technological changes are adopted (Qin et al., 2015).

Water intensity, i.e., water withdrawal or consumption per unit of electricity generation, is spatially heterogeneous because thermal power plants in different areas use various cooling technologies (Macknick et al., 2012). Open-loop cooling plants have a large water withdrawal intensity but a small water consumption intensity. Most of the water withdrawn is discharged to water bodies with high thermal pollution, potentially elevating the water temperature and causing unintended consequences to the aquatic ecosystem, such as reducing the dissolved

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oxygen and killing aquatic wildlife (Madden et al., 2013). On the other hand, closed-loop cooling (recirculating ponds or towers) plants have a much smaller water withdrawal intensity than that of open-loop cooling plants but a larger water consumption intensity. A plant with a recirculating pond may discharge thermal pollution to a water body if the return flow is not completely cooled before being discharged. Recirculating-tower plants have little impact on the water temperature since most of the cooling water evaporates through the tower rather than being discharged into a water body. In general, thermoelectricity supply infrastructures substantially alter the local hydrology and need to be well managed (McManamay et al., 2017).

Environmental regulations involving water withdrawal, consumption, and temperature constraints are commonly adopted to regulate thermoelectric power plants and limit their impacts on public waters. In the US, water withdrawal constraints are implemented per section 316 (b) of the US Clean Water Act (US EPA, 2013) during the planning or construction stages of a plant. These constraints require that the water withdrawal does not exceed the allowable amount to minimize the adverse effects, e.g., entrainment and impingement, on aquatic species associated with the plant's intake. Enforced by the National Pollution Discharge Elimination System (NPDES) (Gaba, 2007) under section 316 (a) of the Clean Water Act, water temperature constraints require that the discharge water does not exceed the allowable maximum temperature or maximum temperature change enacted to protect aquatic ecosystems (Logan and Stillwell, 2018). State or local regulations (IPCB, 2018; Texas Administrative Code, 2014) are also used to further dictate the discharge water temperature. Temperature constraints are applied only to those plants that discharge thermal pollution to surface water. There are no specific federal policies that regulate cooling water consumption in the US, though requests for such policies have been made in recent years (Eheart, 2004; Zhang and Balay, 2014). The absence of water consumption constraints may influence the water supplies used for other purposes during a drought event when a significant amount of water is used for cooling. Cooling water consumption is assumed to be no larger than 1% of the streamflow (Gjorgiev and Sansavini, 2018), which may be underestimated during a drought.

In recent years, many cases of curtailment and shutdown have been reported in Europe (Payet-Burin et al., 2018; Povoledo, 2003), in the US (Averyt et al., 2011; Lydersen, 2016), and in other countries (WRI, 2017; Luo et al., 2018) during droughts because of limited cooling water availability and high water temperatures. These circumstances have led to power outages in some regions, resulting in chaos (BBC, 2003) and even increased mortality (Anderson and Bell, 2012). As a result, thermal variances or relaxed constraints were proposed (Lubega and Stillwell, 2018) or granted (Kopytko and Perkins, 2011; Lydersen, 2016) to power plants during droughts and heat waves to maintain the electricity supply. However, these variances are considered anticompetitive subsidies and are controversial as a way to sustain the electricity supply, even during drought events (Micha, 2014).

To address these issues, some models have been developed (Sattler et al., 2012; Miara and Vörösmarty, 2013; Miara et al., 2013; Jornada and Leon, 2016; Parkinson et al., 2016; Khan et al., 2018) to investigate the water-electricity nexus at a regional or national scale in planning for the future. Sattler et al. (2012) integrates the Regional Energy Deployment System (ReEDS) with the Water Evaluation and Planning (WEAP) model, which is used to investigate the impacts of different future electricity profiles on water resources in the southwestern US (Yates et al., 2013). Miara and Vörösmarty (2013) develops a Thermoelectric Power & Thermal Pollution Model (TP2M) for a hypothetical watershed to examine the tradeoffs between electricity production and thermal pollution. Findings show that strict water constraints may reduce some plants' electricity generation, but may also increase generation efficiency at the watershed scale (Miara et al., 2013). TP2M is applied in Miara et al. (2018) to analyze the impact of thermal pollution on the electricity supply in the Mississippi River watershed. A two-stage optimization model that minimizes capital costs and water withdrawal in

the first stage, and maximizes the robustness under future uncertainties in the second stage, is built in Texas to address electricity expansion and planning problems (Jornada and Leon, 2016). Parkinson et al. (2016) integrates the electricity and freshwater supply systems to ascertain the feasibility of low-carbon electricity strategies in Saudi Arabia under the constraints of a groundwater conservation policy. Khan et al. (2018) develops a coupled water-energy optimization model that tracks the entire life cycle of water and energy with linear representations for long-term resource security enhancement and to increase resource use efficiencies.

In these models, power plants in a watershed or region with similar fuels and cooling technologies are usually aggregated as one representative plant to simplify the simulation (e.g., Sattler et al., 2012). As a result, spatial distributions of power plants and local streamflow conditions are usually overlooked. These models generally consider the long-term water resources conditions, while neglecting the impacts of a specific drought or low flow episode. The models typically provide long-term electricity expansion strategies, such as the technology and location selections of new power plants. For example, retrofitting to water-efficient cooling technologies (Tidwell et al., 2014) is proposed to enhance the reliability of the future electricity supply, regardless of the large capital costs and energy penalties. Some studies, e.g., Khan et al. (2018), consider the cost as one of the criteria when evaluating the reliability of the water-electricity nexus.

Optimization models for the short term (Cortinovis et al., 2009; Sanders et al., 2014; Sanders, 2015; Fooladivanda and Taylor, 2015; Gjorgiev and Sansavini, 2017, 2018; Lubega and Stillwell, 2018) have also been developed at a finer spatial scale, e.g., a watershed scale, to examine the electricity supply reliability during a drought when both water and electricity demands peak and cooling technologies cannot be immediately retrofitted. Cortinovis et al. (2009) presents a systemic approach to optimize the operation of a wet-cooling tower system by minimizing operating costs, including cooling water costs. Sanders et al. (2014) uses a unit commitment and dispatch model, PLEXOS, to estimate how the dispatch and water use of power plants would respond to various levels of water fees in the Electric Reliability Council of Texas (ERCOT), which is responsible for operating the power grid throughout most of the state. Fooladivanda and Taylor (2015) demonstrates a numerical dispatch model to minimize the total cost of a power system while satisfying the constraints, including water use of once-through cooling systems. Gjorgiev and Sansavini (2017) develops a smart water management model that integrates the operation of an upstream hydropower plant and a downstream thermal power plant to maximize the gross electricity output and reduce possible power curtailment of the downstream plant with a once-through cooling system. The model has been applied to investigate the impacts of water constraints on electricity generation (Gjorgiev and Sansavini, 2018). Lubega and Stillwell (2018) develops a linear optimization model to identify how much and to which plants the thermal pollution variances should be granted to maintain the power grid reliability during a specific drought event.

Most of the models investigate the reliability of the open-loop or recirculating-pond plants, which may resort to violating the water temperature constraints for more power generation during a hot summer. Some thermal variances are proposed (Gjorgiev and Sansavini, 2018; Lubega and Stillwell, 2018) to avoid power plant curtailment or temporary shutdown. The impact of the cooling water consumption on streamflow is usually neither considered nor underestimated in previous studies. This impact is minimal during normal hydrologic conditions or wet seasons but can be serious during a drought event, especially in a watershed with many closed-loop cooling plants that consume a significant amount of water. Cooling technologies are shifting from open-loop to closed-loop, which would exert increasing effects on water consumption but decreasing impacts on water withdrawal and temperature. The existing models usually oversimplify hydrologic processes and use observed streamflow or results from conceptual hydrologic models to represent the cooling water availability. For example,

Gjorgiev and Sansavini (2018) use streamflow and water temperature as exogenous inputs but do not explicitly simulate them in the modeling framework. Lubega and Stillwell (2018) use principal components analysis (PCA) to determine streamflow. Thus, the dynamics of streamflow and the impact of upstream cooling water consumption on downstream water availability and power generation cannot be simulated in detail, which, to some degree, limits the applicability of these models to ungagged sites and future scenarios. In addition, the top-down centralized optimization framework commonly used in the existing models assumes full cooperation among the power plants to achieve watershed-wide benefits, which is usually not true in the real world and would overestimate the potential electricity supply (Gjorgiev and Sansavini, 2018). The centralized optimization framework approaches the

realistic representation when all plants in a region are operated by the same utility. In other cases, plants would maximize their own profits rather than fully cooperate with each other to achieve system-wide benefits. Therefore, a decentralized optimization framework (Yang et al., 2009) is preferred in this context, as in our study.

This study contributes to water-energy nexus research communities by providing a coupled modeling tool, PyHSPF-WE (Water-Electricity nexus model based on a hydrologic model Hydrological Simulation Program-Fortran written in Python programming language), closely combining hydrologic processes (streamflow and water temperature), thermoelectricity generation, and economic incentive determination at the watershed level and at the daily time step. First, this tool enables a more explicit assessment of the impact of thermal power plant cooling

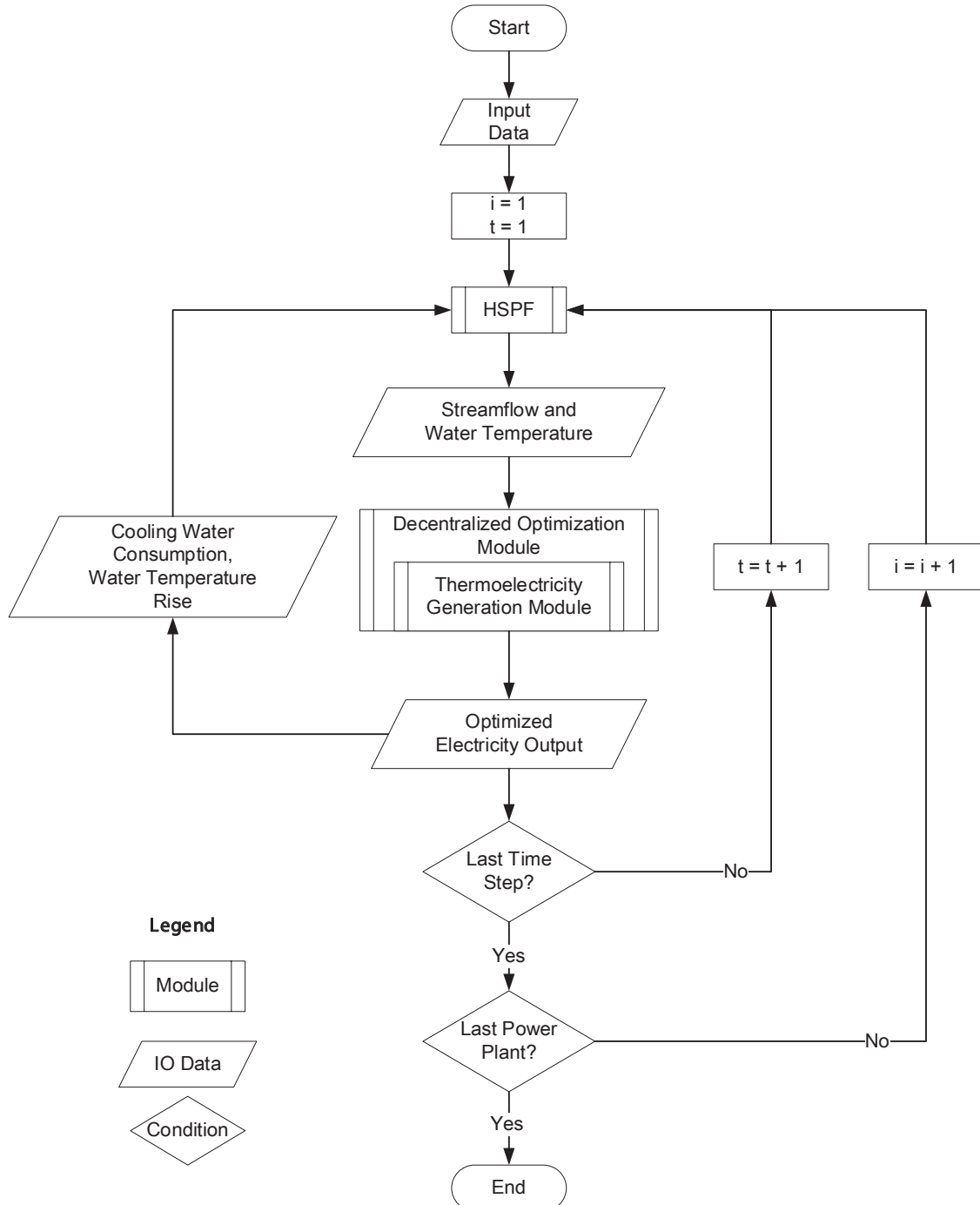


Fig. 1. Workflow of the water-electricity nexus model. (t : time index; i : plant index.)

water consumption on daily streamflow, which is commonly under-represented in previous studies (e.g., Gjorgiev and Sansavini, 2018; Lubega and Stillwell, 2018). Second, solutions that involve both environmental regulations (i.e., via different levels of environmental constraints) and economic penalties are explored to deal with the tradeoffs between electricity generation and environment protection. Third, spatially distinguished policies are examined for different thermal power plants, thereby producing more realistic solutions that lead to more environmental benefits and electricity generation within a watershed.

The physically based hydrologic module, Hydrological Simulation Program-Fortran (HSPF) (Bicknell et al., 2005), is used to simulate daily hydrological processes and take into account the impact of cooling water consumption on streamflow and power generation over time. The thermoelectricity generation module links cooling water consumption and thermal pollution to the electricity generation of a plant. We formulate a decentralized optimization module (Yang et al., 2009) for each power plant to determine how power plants would respond to different hydrologic conditions, environmental constraints, and economic penalties. A case study is conducted for the drought in 2012 (Knapp et al., 2017) in the Kaskaskia River watershed in Illinois, US to demonstrate the model applicability, while exploring the tradeoffs between environmental constraints and the electricity supply. The potential for using economic penalties to mitigate the environmental effects of electricity generation and the upstream vs. downstream effect at the watershed level is also demonstrated. This water-electricity nexus model can be easily adopted for other watersheds to examine the thermoelectricity supply reliability under environmental regulations and economic penalties, serving as a versatile modeling tool to investigate a wide range of policy issues regarding the water-electricity nexus.

2. Methods

The flowchart in Fig. 1 shows the workflow for integrating three modules to create a water-electricity nexus model. An interface is developed that integrates these three modules on a daily basis. At each time step, the HSPF module assimilates the necessary input data and simulates the hydrologic conditions, daily streamflow and water temperature, that serve as the input for the other two modules. The decentralized optimization module is then used to optimize the power plant's electricity generation to maximize its own profits. The electricity generation of each plant that has a specific cooling technology further determines the cooling water consumption, which would be input back to HSPF to update the initial hydrologic conditions of the next time step. These processes iterate over time and space, i.e., from upstream to downstream plants, to determine the impact of cooling water consumption on the next-time-step and downstream hydrologic conditions. This nexus modeling framework captures the interactions between hydrology and thermoelectricity generation. The following is the detailed description of each module.

2.1. The hydrologic module: HSPF

HSPF (Bicknell et al., 2005) is a physically based hydrologic model that can simulate the spatially distributed water quantity and quality processes at a watershed scale and at various time steps from 1 min to one year. The model has been developed and maintained by the US Environmental Protection Agency (EPA) and the US Geological Survey (USGS) since the 1960s. In the HSPF, a watershed is divided into individual Hydrologic Response Units (HRUs), in which the hydrologic responses are assumed homogeneous. HSPF uses climate forcing, land use, and hydrography data as inputs and provides hydrologic variables, such as streamflow and river water temperature, as outputs.

With the original FORTRAN codes, the HSPF is time-consuming to compile, maintain, and extend. For example, the most recent model release does not provide formal makefiles or detailed documentation to

compile the source codes. Here, we use a recently developed open-source HSPF interface called PyHSPF (Lampert and Wu, 2015, 2018) to prepare the input data and run the HSPF. Coded in the Python programming language, PyHSPF offers the advantages of having an open-source nature and the flexibility to extend for certain purposes. In our study, we further extend the PyHSPF by 1) adding a new interface to interact with the water temperature simulation module in the HSPF, and 2) adding a new interface to account for the impact of cooling water consumption on streamflow. HSPF is calibrated and run at a daily time step using the PyHSPF. The extended PyHSPF is further integrated with the decentralized optimization and thermoelectricity generation modules.

2.2. The decentralized optimization module

The optimization problem in this study is formulated in a decentralized optimization framework (Yang et al., 2009) to determine how much electricity would be produced by each plant in response to specific streamflow conditions, environmental regulations, and economic penalties. This decentralized optimization can represent a more realistic setting taken from the real world, in which the upstream and downstream plants usually do not cooperate with each other to achieve the best system-level benefit, but instead manage to maximize their individual benefits, i.e., the economic profit in this context. This setting indicates that an upstream plant would use as much water and discharge as much thermal pollution as necessary to maximize its profit as long as the environmental constraints are not violated. The upstream power plants would not sacrifice their water rights to conserve water for those downstream. The result of the decentralized optimization would converge with that of the centralized optimization if the upstream plants are always more water-efficient than those downstream when a water shortage occurs during a drought (Yang et al., 2009). The following describes the decentralized optimization formulation used in this study.

The optimization of each plant is conducted from upstream to downstream at each daily time step to maximize its own objective function $f_{i,t}$:

$$\max_{e_{i,t}} f_{i,t}(e_{i,t}, Q_{i,t}, T_{i,t}^{\text{in}}) \quad (1)$$

where i is the index of the plants, t is the time index, $e_{i,t}$ is the daily electricity generation (MWh), which is the decision variable, and $Q_{i,t}$ and $T_{i,t}^{\text{in}}$ are the streamflow (m^3/s) and inlet water temperature ($^{\circ}\text{C}$), respectively.

The above optimization problem is limited by four constraints for each plant i . The first is the physical limitation of the generating unit of the plant:

$$E_i^{\text{min}} \leq e_{i,t}/24 \leq E_i^{\text{max}} \text{ or } e_{i,t} = 0 \quad (2)$$

where E_i^{min} and E_i^{max} are the minimum load (MW) and maximum capacity (MW). Power plants can operate continuously only above the minimum load and below the maximum capacity. Otherwise, plants will temporarily shut down, thereby reducing their electricity output to zero.

The second constraint is introduced by the cooling water consumption rate constraint and applies to all plants, regardless of their cooling technologies:

$$W_{i,t} \leq (W_0 + \beta_w) Q_{i,t} \quad (3)$$

where $W_{i,t}$ is the cooling water consumption rate (m^3/s) and W_0 is the initial water consumption rate constraint, i.e., 1% (Gjorgiev and Sansavini, 2018). β_w is the constraint relaxation, which varies from 0 to 14% at the step of 1.4% in this study, used to analyze the tradeoffs between water consumption and the electricity supply. A maximum constraint relaxation of 14% allows operating all the plants at full capacity during the studied period.

The other two constraints are introduced by water temperature regulations, which apply to plants with a cooling pond and not to those with recirculating towers:

$$T_{i,t}^{out} \leq T0 + \beta_T \quad (4)$$

$$\Delta T_{i,t} \leq \Delta T0 + \beta_T \quad (5)$$

where $T0$ and $\Delta T0$ are the initial allowable maximum temperature and temperature rise above the natural water temperature ($^{\circ}\text{C}$), and β_T is the constraint relaxation for thermal pollution, varying from 0 to 3°C at the step of 0.3°C in this study. A maximum temperature constraint relaxation of 3°C allows operating all the temperature-sensitive plants at full capacity during the studied period.

The average electricity output (E_i : MW) by plant i during the summer months can be calculated as:

$$E_i = \frac{1}{24 * N_t} \sum_t e_{i,t} \quad (6)$$

where N_t is the length of the study period, i.e., 92 days in the case study.

The total electricity output (MW) at the watershed level is the sum of the electricity output by all plants located in the study watershed over time:

$$\sum_i E_i = \frac{1}{24 * N_t} \sum_{i,t} e_{i,t} \quad (7)$$

2.3. The thermoelectricity generation module

The following thermoelectricity generation module links the HSPF and the optimization modules by defining the objective function and relevant components, such as the water consumption requirement and thermal pollution. The objective function f is defined as the daily profit of the plant:

$$f_{i,t} = R_{i,t} - OM_{i,t} - PW_{i,t} - PT_{i,t} \quad (8)$$

where i is the plant index, t is the time index, $R_{i,t}$ is the revenue (\$) for selling electricity to the market, $OM_{i,t}$ is the operating and maintenance cost (\$) (including the fuel cost), and $PW_{i,t}$ and $PT_{i,t}$ are the economic penalties (\$) for the extra water consumption and water temperature rise.

The revenue for selling electricity to the market is:

$$R_{i,t} = e_{i,t} * EP_t \quad (9)$$

where $e_{i,t}$ is the daily electricity generation (MWh) and EP_t is the electricity price (\$/MWh) in the market.

The O&M cost is:

$$OM_{i,t} = e_{i,t} * OP_i \quad (10)$$

where OP_i is the plant-specific O&M cost per unit of electricity generation (\$/MWh).

In the following pages, we introduce the concepts of economic penalties for plants that violate the initial environmental constraints, $W0$, $T0$, and $\Delta T0$. During a drought, some policy flexibility and constraints relaxation are usually assumed necessary (Gjorgiev and Sansavini, 2018; Lubega and Stillwell, 2018) and are granted by environmental agencies. In this study, we propose that economic penalties should also be imposed on the plants that relax the constraints, since a water pricing strategy has been advocated to have more advantages and social welfare gains compared to those of the policy controls (Mansur and Olmstead, 2012; Luby et al., 2018). In the short run, these penalties may result in a greater contribution to the electricity market by water-efficient plants and a smaller contribution by water-inefficient plants. The penalties that the plants pay may be used for environmental recovery after the drought. In the long run, these

penalties may encourage plants to retrofit to more water-efficient cooling technologies to avoid economic penalties, eventually reducing the impacts on water resources.

The penalty for extra water consumption (PW) is calculated as:

$$PW_{i,t} = \alpha_W \max(W_{i,t} - W0 * Q_{i,t}, 0)^2 \quad (11)$$

where $W0$ is the initial water consumption constraint, which is set as 1% (Gjorgiev and Sansavini, 2018) due to the absence of general policies and regulations related to cooling water consumption in the US (Eheart, 2004; Zhang and Balay, 2014). $W0 * Q_{i,t}$ is the allowable maximum cooling water consumption rate (m^3/s) if the initial water consumption policy is strictly enforced and when the consumption rate of cooling water does not exceed 1% of the natural streamflow $Q_{i,t}$. In this context, $\max(W_{i,t} - W0 * Q_{i,t}, 0)$ is the daily average extra water consumption rate (m^3/s); α_W is the penalty factor (\$/(\$/\text{m}^3/\text{s})^2) for violating the water consumption rate constraint; and $\alpha_W * \max(W_{i,t} - W0 * Q_{i,t}, 0)$ is the penalty price (\$/(\$/\text{m}^3/\text{s})) of the extra water consumption, which means the plants would have to pay a higher unit price for a higher extra water consumption rate. The penalty price equals zero and the plants do not need to pay if water consumption $W_{i,t}$ is less than 1% ($W0$) of $Q_{i,t}$ when the initial water consumption constraint is not violated.

The cooling water consumption rate ($W_{i,t}$: m^3/s) is determined by electricity generation and water consumption intensity:

$$W_{i,t} = e_{i,t} * WI_i / (3600 * 24) \quad (12)$$

where WI_i is the plant-specific cooling water consumption intensity (m^3/MWh) and $3600 * 24$ is the unit conversion factor.

Similarly, the penalty for an extra temperature rise (PT) is defined as:

$$PT_{i,t} = \alpha_T \max(T_{i,t}^{out} - T0, \Delta T_{i,t} - \Delta T0, 0)^2 \quad (13)$$

where $T0$ and $\Delta T0$ are the allowable maximum temperature and temperature rise above the natural water temperature in the initial policy context. $T_{i,t}^{out}$ is the water temperature at the ending point of the cooling mixing zone. $\Delta T_{i,t}$ is the temperature rise caused by thermal pollution, which elevates the water temperature from $T_{i,t}^{\text{in}}$ to $T_{i,t}^{\text{out}}$. α_T and $\alpha_T * \max(T_{i,t}^{out} - T0, \Delta T_{i,t} - \Delta T0, 0)$ are the penalty factor (\$/(\$/\text{C})^2) and penalty price (\$/(\$/\text{C})), respectively, for violating the initial water temperature constraints. The plants would not be penalized if none of the temperature constraints were violated. $PT_{i,t}$ is applied only to recirculating-pond cooling plants in this study since recirculating-tower cooling plants discharge minimal thermal pollution to the river.

The water temperature rise ($\Delta T_{i,t}$) is calculated following the heat transfer model and energy balance:

$$\Delta T_{i,t} = \frac{e_{i,t} (1 - \eta_i)}{\eta_i} / (V_i * \rho * C_p) * 3.6 * 10^9 \quad (14)$$

where η_i is the thermal efficiency of the plant (%), which is assumed constant during the study period (Gjorgiev and Sansavini, 2017, 2018; Lubega and Stillwell, 2018), although it actually fluctuates with the climatic and hydrologic conditions and the performance of generators. V_i is the volume of the cooling mixing zone (m^3). ρ is the water density (kg/m^3). C_p is the heat capacity of water ($\text{J}/\text{kg}/^{\circ}\text{C}$). $3.6 * 10^9$ is the conversion factor that converts the unit MWh to J.

The outlet water temperature ($T_{i,t}^{\text{out}}$) at the cooling mixing zone is:

$$T_{i,t}^{\text{out}} = T_{i,t}^{\text{in}} + \Delta T_{i,t} \quad (15)$$

where $T_{i,t}^{\text{in}}$ is the inlet river water temperature.

The decentralized optimization module is formulated in the Pyomo (Hart et al., 2017) programming environment and solved with a nonlinear solver IPOPT (Wächter and Biegler, 2006), which allows high-speed computing with good accuracy. A closed-loop optimization

is performed in terms of the feedback between the decisions related to electricity generation, the environmental impacts, and policy controls. One sub-optimization process is conducted for each power plant during the study period, so the number of sub-optimization processes depends on the number of plants in the case study.

3. Case study and data

3.1. Case study description

A case study is conducted in the Kaskaskia River watershed (Fig. 2) in southern Illinois to demonstrate the application of the water-electricity nexus model. This watershed, which hosts four large thermal power plants (EIA, 2016a), denoted by P1–P4, suffered from the 2012 drought (Knapp et al., 2017). Many communities in this region were forced to limit their water usage (Erickson, 2012), while the power plants consumed a significant amount of cooling water during this period to generate electricity and discharged thermal effluents at high temperatures (EIA, 2016b). Four 8-digit HUCs (Hydrologic Unit Codes) are contained in this watershed and are divided into 106 HRUs (Hydrologic Response Units). The study period focuses on the drought event in summer 2012 (92 days from June 1 to August 31) (Miara et al., 2018) when the electricity demands were at the peak, the water temperatures were the highest, and the water flows were the lowest. Power plants P1 and P2 are located upstream, and thus their cooling water availabilities are not affected by other plants. The nexus model runs at P1 and P2 over time to determine the dynamics of cooling water availability and electricity generation. The impact of the current-time-step cooling water consumption on the next-time-step cooling water availability is included at each plant. After finishing the modeling processes at P1 and P2, the nexus model moves forward to P3 while considering the impact of P1 and P2, and then to P4 while considering the impact of all three upstream plants. In this way, the impact of the upstream plants' cooling water consumption on downstream plants is included.

Since currently there is no specific cooling water consumption constraint applied in this watershed, 1% (Gjorgiev and Sansavini, 2018)

of the natural streamflow at each plant is set as the initial water consumption constraint to limit the impact on water resources. Required by Code 302.211 in Title 35 of the Illinois Administrative Code (IPCB, 2018), the allowable maximum temperature (T_0) and temperature rise (ΔT_0) above the natural water temperature at the cooling mixing zone should not exceed 32°C and 2.8°C, respectively. In the model, relaxations (β_W and β_T) of water consumption constraints and temperature constraints are granted to plants when necessary during a drought to maintain the reliability of the electricity supply, but with subsequent economic penalties (α_W and α_T).

The water-electricity nexus model built in the case study is implemented on a PC with an Intel (R) Core (TM) i5-2500 CPU (3.30 GHz). Each run with a specific combination of parameters takes about 25 minutes.

3.2. Input data

Hydrography data in the Kaskaskia River watershed, e.g., flowlines, catchments, elevations, slopes, and boundaries, are extracted from the National Hydrography Dataset Plus Version 2 (NHDPlusV2) (McKay et al., 2012). Land use data are obtained from the Cropland Data Layer (CDL) (NASS, 2017). Streamflow and river water temperature data are downloaded from the USGS National Water Information System (NWIS) (USGS, 2018) for model calibration and validation.

Climate forcing data are obtained from various sources. Station-based precipitations are extracted from the Hourly Precipitation Dataset DSI-3240 (NCDC, 2003) and aggregated for each sub-watershed using the inverse distance weighting method. Daily maximum and minimum temperatures, snow depths, and snowfalls are extracted from the Global Historical Climatology Network (GHCN) (Menne et al., 2016). Daily wind speeds and dew points are extracted from the Global Surface Summary of the Day (GSOD) (NCDC, 2017). Solar radiation data are taken from the National Solar Radiation Database (NSRDB) (Sengupta et al., 2014). Daily cloud cover data are necessary for the water temperature simulation module and are extracted from the Advanced Very High Resolution Radiometer (AVHRR) remote-sensing product PATMOS-x (Heidinger et al., 2014). A summary of the input data for the HSPF is shown in Table 1.

Using the input data in Table 1, the HSPF is run at a daily time step from 1980 to 1995 for streamflow calibration and from 2001 to 2012 for streamflow validation at the gage station USGS 05594100. The simulation between 1995 and 2000 is eliminated because of the inconsistency in precipitation and streamflow observations (high streamflows with low precipitations). The water temperature simulation module in

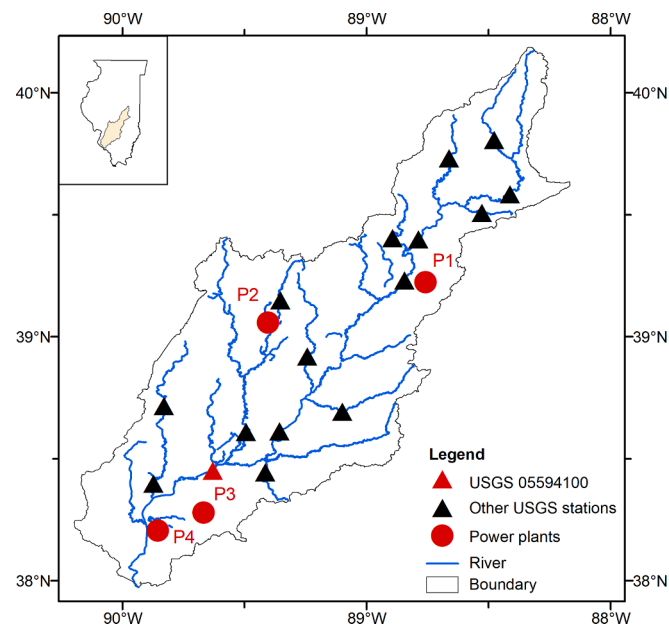


Fig. 2. Kaskaskia River watershed and the locations of four thermal power plants and USGS gage stations. (Daily streamflow observations are provided by the gage station USGS 05594100 for calibration during 1980–1995 and for validation during 2001–2012. Water temperature observations during 1980–1995 are provided by the USGS 05594100 for calibration and by 15 other USGS stations for validation.)

Table 1
Summary of the input data for the hydrologic module HSPF.

Input data	Sources	References
Hydrography data	National Hydrography Dataset Plus Version 2 (NHDPlusV2)	McKay et al., 2012
Land use	Cropland Data Layer (CDL)	NASS, 2017
Streamflow and river water temperature observation	National Water Information System (NWIS)	USGS, 2018
Precipitation	Hourly Precipitation Dataset DSI-3240	NCDC, 2003
Daily maximum and minimum temperatures, snow depths and snowfalls	Global Historical Climatology Network (GHCN)	Menne et al., 2016
Daily wind speeds and dew points	Global Surface Summary of the Day (GSOD)	NCDC, 2017
Solar radiation	National Solar Radiation Database (NSRDB)	Sengupta et al., 2014
Cloud cover	AVHRR PATMOS-x	Heidinger et al., 2014

HSPF is calibrated at USGS 05594100 during 1980–1995 and validated at the other 15 USGS stations during the same period. This calibration method is applicable since all the sub-watersheds share the same calibrated parameters in the water temperature simulation module. The detailed calibration and validation results for streamflow and water temperature can be found in Appendix A. Next, we focus our study period on the 2012 summer: 92 days from June 1 to August 31. The HSPF outputs daily streamflow and water temperature data that serve as the input for the thermoelectricity generation and optimization modules.

The operating and maintenance cost (OP_i) is obtained from the Electric Power Annual Table 8.4 (EIA, 2017a). P1 uses the Natural Gas Fired Combined Cycle (NGCC) and its OP is 35.67 \$/MWh. P2, P3, and P4 use Conventional Steam Coal (CSC) and their OP is 37.60 \$/MWh. Here, the OP differences between P2, P3, and P4 are ignored due to the lack of more detailed information. The plant-specific operating cost is usually confidential and unavailable (Payet-Burin et al., 2018). Average cooling water consumption intensity (WI_i) in summer months is compiled from the US Energy Information Administration (EIA) Thermoelectric Cooling Water Data (EIA, 2017b) for each plant. The minimum load (E_i^{min}) and maximum capacity (E_i^{max}) of each plant are obtained from Form EIA-860 (EIA, 2016a). Thermal efficiency (η_i) of the plant during the summer season is calculated as the average rate of the net electricity output to electric fuel consumption from Form EIA-923 (EIA, 2016b). V_i is the volume of the cooling mixing zone of a recirculating-pond cooling plant, which is obtained from Form EIA-923 (EIA, 2016b). A summary of the characteristic data of the four plants in the Kaskaskia River watershed is shown in Table 2.

The price of electricity during the study period is obtained from the Intercontinental Exchange (ICE) database (ICE, 2012) during the summer of 2012 in the Indiana Hub, where the Kaskaskia River watershed is located.

4. Results

4.1. Electricity supply under varying environmental constraint relaxation levels but no economic penalties

The model results allow us to understand how the electricity supplies at both the plant and watershed levels respond to varying environmental constraint relaxation levels during the summer 2012. For the modeling analysis in this section, economic penalties α_W and α_T for water consumption and temperature constraints violation, respectively, are set as zero, which means the plants are granted constraints relaxation with no penalties. β_T varies from 0 to 3°C at the step of 0.3, and β_W varies from 0 to 14% at the step of 1.4%.

In Fig. 3, the four plants show different responses to the changes of β_W and β_T . The warmer color in Fig. 3 represents a larger value at the Z-axis, and the cooler color represents a smaller value; this same labeling applies to the figures hereinafter if not specified. The electricity supplies by P1 (Fig. 3a) and P3 (Fig. 3c) increase with β_W but are not affected by the increase of β_T since they are equipped with recirculating-tower

cooling technology, which is not limited by temperature constraints. More water consumption constraint relaxation is needed by P1 (14%) than P3 (10%) to reach maximum capacity. P2 (Fig. 3b) uses cooling pond technology and thus is subject to temperature constraints. We assume that P2 is not limited by water consumption constraints since it withdraws water from a nearby reservoir, which can provide adequate cooling water. The minimum temperature constraint relaxation β_T is 1°C for P2 to operate at full capacity. P4 (Fig. 3d) is significantly subject to both water consumption and temperature constraints. When β_W is set as zero, the increase of β_T does not increase the electricity supply at all. When β_T is set as zero, as much as 52% of the capacity can be used when β_W is increased to 14%. The relaxation of β_T is necessary if more than 52% of the capacity is expected by the electricity market. The minimum β_T and β_W are 3°C and 14%, respectively, for P4 to operate at its full capacity. At the watershed level, the sensitivities of the total electricity supplies by the four plants are shown in Fig. 3e. The violation and relaxation of the initial water consumption constraint are necessary if a significant increase in the electricity supply is expected. A larger increase can also be achieved with the relaxation of water temperature constraints. It should be noted that the pattern at the watershed level (Fig. 3e) follows P4 (Fig. 3d). The reason is that P4 has a large capacity and is constrained by both the water consumption and temperature limitations, which is similar to the setting at the watershed level, while the other three power plants are constrained by either consumption or temperature limitations.

4.2. Electricity supply under fixed environmental constraint relaxation levels and varying economic penalties

In this section, the fixed maximum constraint relaxation levels, i.e., $\beta_W = 14\%$ and $\beta_T = 3^\circ\text{C}$, are set for the four plants but with varying economic penalties. The penalty factor α_W (\$/(m³/s)²) for extra water consumption varies from 0 to 10 at the step of 1, and α_T (\$/(°C)²) for an extra temperature rise varies from 0 to 2 at the step of 0.2. Fig. 4 shows the response of electricity supplies to varying economic penalties and the associated cooling water consumption.

The electricity supplies by P1 (Fig. 4a) and P3 (Fig. 4c) decrease with an increase of α_W but are not affected by α_T since they are insensitive to temperature constraints. More than half of their capacities will be curtailed with the maximum α_W as 10. P2 (Fig. 4b) will decrease the electricity supply when α_T increases. Of its capacity, 20% will be curtailed if the maximum α_T is imposed. The electricity supply by P4 (Fig. 4d) will experience a quick drop, even if a small α_W and α_T are adopted. In some cases, P4 would slightly increase its electricity supply when the penalty factor increases, for example, when α_W is fixed as 4 and α_T increases from 0 to 0.2. This result is reasonable since P4 is located downstream of the Kaskaskia River and can receive more cooling water if upstream plants use less water when a penalty is imposed. However, when the penalty factor becomes larger, P4 would choose to decrease its electricity supply to avoid a penalty and maximize its profits, even though there is more available cooling water. The total electricity supply (Fig. 4e) at the watershed level closely follows the pattern of P4.

The second Z-axis in Fig. 4 demonstrates the cooling water consumption (W). The patterns of W exactly follow the patterns of E , since W is a linear function of E , as shown in equation (12) in Section 2.3. In Fig. 4, generally, the downstream plants P3 and P4 (Fig. 4c and Fig. 4d) consume more water than the upstream plants P1 and P2 do (Fig. 4a and Fig. 4b). This is a reasonable location design since power plants with a large capacity and water consumption, such as P3 and P4, are usually located downstream where the water is adequate to support electricity production except during a drought. Comparing W in Fig. 4c and Fig. 4d, P4 consumes nearly 1.5 times as much as P3 with a same level of electricity production, which is caused by the larger cooling water consumption intensity at P4.

Another aspect of the environmental impacts, the rise in water

Table 2

Characteristic data of the four thermal power plants used in this study. Data sources: EIA, 2017a; EIA, 2017b; EIA, 2016a; EIA, 2016b. (i: plant index, E_i^{max} : maximum capacity (MW), E_i^{min} : minimum load (MW), WI_i : water consumption intensity (m³/MWh), η_i : thermal efficiency (%), V_i : volume of cooling mixing zone (10⁶m³), OP_i : maintenance and operating cost (\$/MWh)).

i	Primary Fuel	E_i^{max}	E_i^{min}	Cooling Technology	WI_i	η_i	V_i	OP_i
1	Natural Gas	610	50	Tower	2.67	45	-	35.7
2	Coal	895	190	Pond	0.90	31	27.1	37.6
3	Coal	1628	420	Tower	1.62	36	-	37.6
4	Coal	1778	250	Pond	2.48	33	33.3	37.6

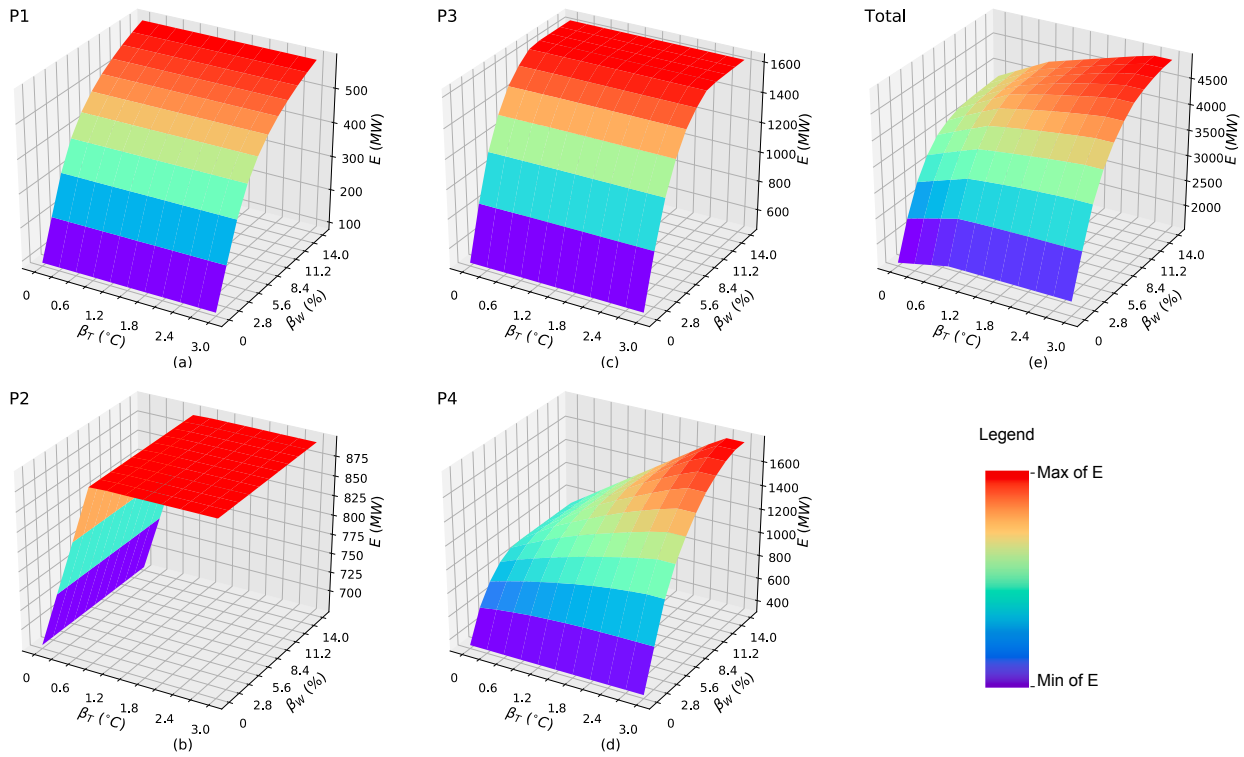


Fig. 3. Electricity supply (E) by (a) P1, (b) P2, (c) P3, (d) P4, and (e) total four plants, under varying water consumption (β_W) and temperature (β_T) constraint relaxation levels but no economic penalties ($\alpha_W = \alpha_T = 0$). The warmer color represents a larger value at the Z-axis, and the cooler color represents a smaller value; the same labeling applies to the figures hereinafter if not specified.

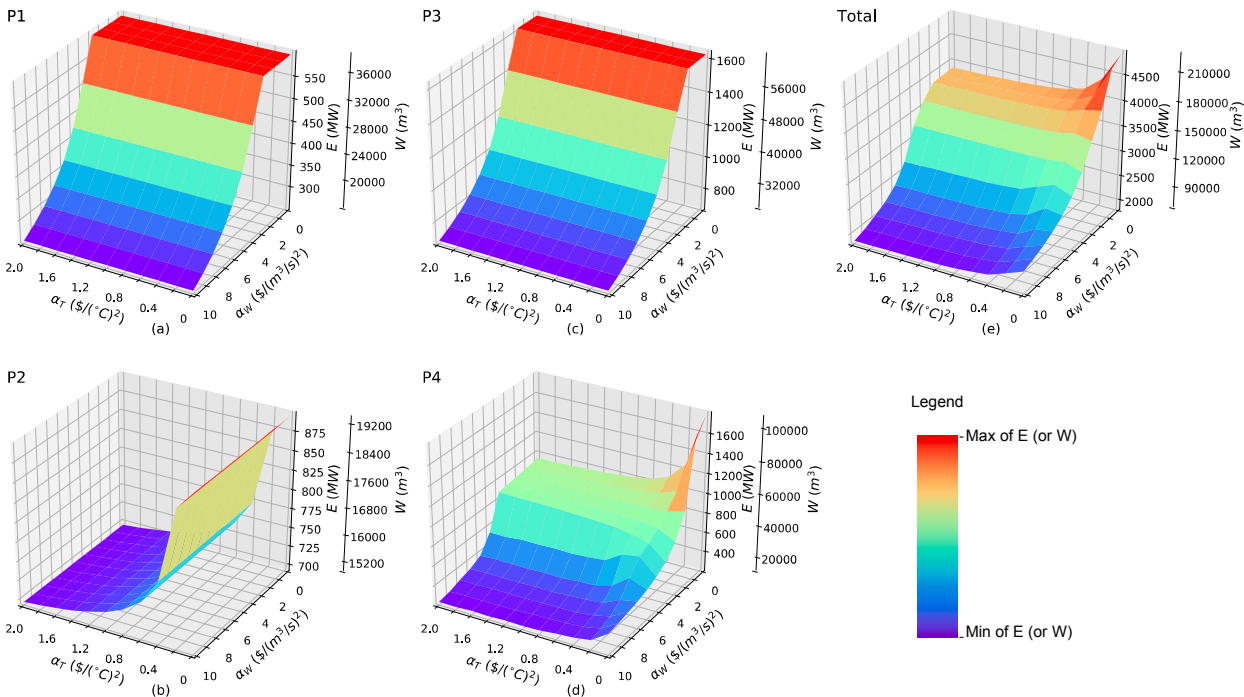


Fig. 4. Electricity supply (E) and cooling water consumption (W) by (a) P1, (b) P2, (c) P3, (d) P4, and (e) total four plants, under fixed environmental constraint relaxation levels for water consumption ($\beta_W = 14\%$) and temperature ($\beta_T = 3^\circ\text{C}$), and varying economic penalty factors for violation of water consumption constraints (α_W) and temperature constraints (α_T).

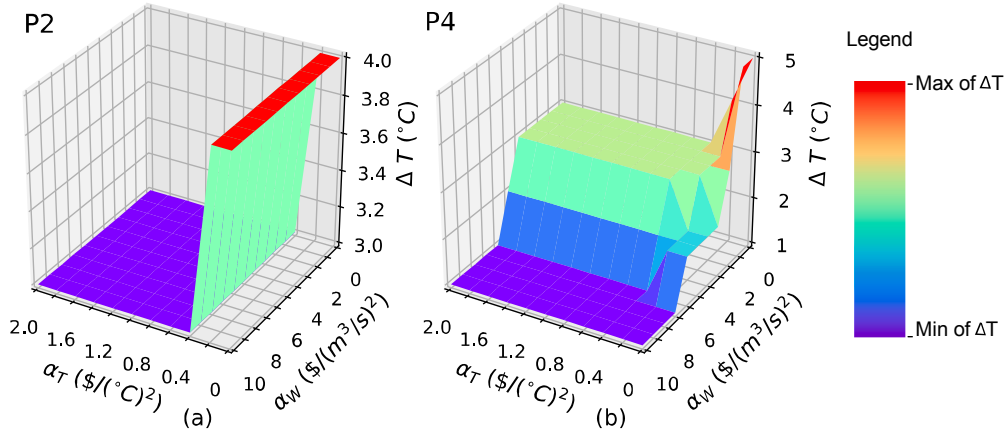


Fig. 5. Water temperature rise (ΔT) by power plants (a) P2 and (b) P4. (P1 and P3 do not discharge thermal pollution to water bodies and raise the water temperature.)

temperature (ΔT) by plants P2 and P4, is quantified in Fig. 5. P1 and P3 do not discharge thermal pollution to water bodies, raising the water temperature. Focusing on the α_T axis and fixing α_W as zero, when the penalty factor on a water temperature rise is larger than 0.4, both P2 and P4 will reduce ΔT to around the initial allowable maximum level of 2.8°C to avoid economic losses. However, in Fig. 4b and Fig. 4d, the electricity generation will continue to slightly decrease with a larger α_T . This implicates that adopting an α_T larger than 0.4 is impractical when making the policy decision since it does not significantly reduce ΔT but decreases the electricity generation. At P4 in Fig. 5b, ΔT is further reduced to lower than 2.8°C when α_W is increased, which indicates that a larger economic penalty on the extra water consumption would also help to relieve the environmental impacts of a rising water temperature. P2 is not limited by the cooling water constraint β_W or the economic penalty α_W .

4.3. Electricity supply under varying environmental constraint relaxation levels and economic penalties

Following the results from the modeling exercises that have fixed environmental constraint relaxation levels or economic penalties to separate their impacts on electricity supply, in this section, different combinations of constraint relaxation levels and economic penalties are used to investigate the joint impacts at the watershed level.

Economic penalties are set as five levels in the subfigures of Fig. 6, as α_W increases from 0 to 10 and α_T increases from 0 to 2. In each subfigure, the increase of β_W at the step of 1.4% is represented by the horizontal axis, and the increase of β_T at the step of 0.2 is represented by scenarios S1 to S11. When no penalties are applied (Fig. 6a), the electricity supply

increases with β_W and β_T , though at different rates. The increase of E is largely driven by β_W when β_W is smaller than 5.6%, and the increase of β_T contributes slightly to the increase of E . When β_W is larger than 8.4%, the increase of E is largely driven by β_T and the increase of β_W contributes just a little. Different combinations of β_W and β_T can maintain the same level of electricity supply, e.g., $E_A = E_B$ (Fig. 6a). Comparing A to B, there is less impact on water quantity but more on water temperature. To increase the electricity supply from E_B to E_C (or E_D), we can choose the path from B to D, which consumes more water but with the same level of impact on water temperature, or the path from B to C, which causes a greater water temperature rise but with the same water consumption. These tradeoffs between the electricity supply, water consumption, and river water temperature need to be balanced by watershed agencies during a drought event. When larger penalties are imposed (from Fig. 6a to Fig. 6e), profit-driven plants would decrease their electricity supplies to avoid paying a penalty. Comparing Fig. 6b to Fig. 6a, a small increase of α_T would cause a quick decrease of E and reduce the differences among scenarios S1 to S11, which means that the temperature constraints relaxation cannot encourage an increased electricity supply if the penalty factor α_T is larger than 0.25. The contribution to E by the increase of β_W would significantly decrease if the penalty factor α_W is larger than 5. In Fig. 6e, E would not increase with β_W when β_W is larger than 1.4% since the penalty is high for violating the initial cooling water consumption constraint.

Generally, the upstream aquatic environment in a watershed is more fragile and sensitive than the one downstream. The water availability is usually smaller upstream and its capability to absorb the hydrologic disturbance is limited. In addition, upstream water consumption would result in a cumulative impact on the aquatic ecosystem and less water

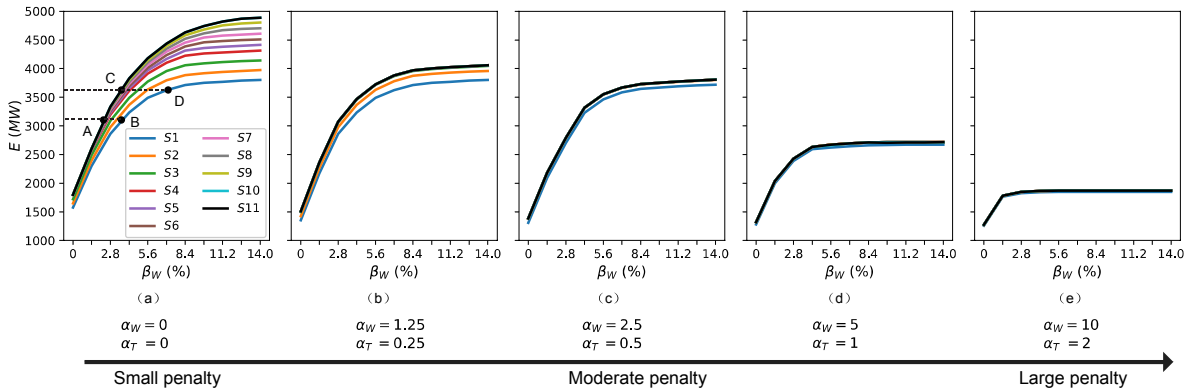


Fig. 6. Watershed-level total electricity supply (E) under varying water consumption (β_W) and temperature (β_T) constraint relaxation levels and economic penalties (α_W and α_T). Scenarios S1 to S11 represent β_T increasing from 0°C to 2°C at the step of 0.2°C.

availability for users downstream. Therefore, resource agencies may apply more stringent constraints or penalties at the headwaters than at downstream locations (SRBC, 2012). For this reason, we further apply a spatially distinguished penalty, or penalty price discrimination (Stole, 2007), by imposing a larger penalty factor on the two upstream plants (P1 and P2) and a smaller factor on the two downstream plants (P3 and P4). Ten levels of penalty price discrimination are applied. The penalty factor for extra cooling water consumption under the price discrimination is:

$$\alpha'_w = \begin{cases} \alpha_w * (1 + p), & i = 1, 2 \\ \alpha_w * (1 - p), & i = 3, 4 \end{cases} \quad (16)$$

The penalty factor for an extra rise in water temperature under the discrimination is:

$$\alpha'_T = \begin{cases} \alpha_T * (1 + p), & i = 1, 2 \\ \alpha_T * (1 - p), & i = 3, 4 \end{cases} \quad (17)$$

where p is the penalty discrimination level expressed as a percentage, varying from 0 to 100%. When $p = 0$, all four plants share the same penalty factors $\alpha_w = 5$ and $\alpha_T = 1$, which are the same as in Fig. 6d, regarded as the baseline penalty scenario. A positive p means a larger penalty for upstream plants and a smaller penalty for downstream plants. When $p = 100\%$, the largest penalty is imposed on the upstream plants, but there is no penalty for those downstream. Scenarios S'1 to S'11 represent the different levels of environmental constraint relaxation, with β_T increasing from 0°C to 2°C at the step of 0.2°C and β_w increasing from 0 to 14% at the step of 1.4%. As shown in Fig. 7, the watershed-level total electricity supply would significantly increase if the penalty discrimination is applied in this watershed. This kind of increase shows in all constraint relaxation scenarios, S'1 to S'11. Some small decreases are observed when p varies from 90% to 100% in scenarios S'2 to S'5. Compared to the baseline penalty scenario, as much as 60% more electricity can be supplied if the largest penalty discrimination is applied in scenario S'11 with the maximum constraints relaxation.

These modeling results reveal the quantitative tradeoffs between the electricity supply and environmental constraints under various management regulations and economic penalties. The final decision is left to

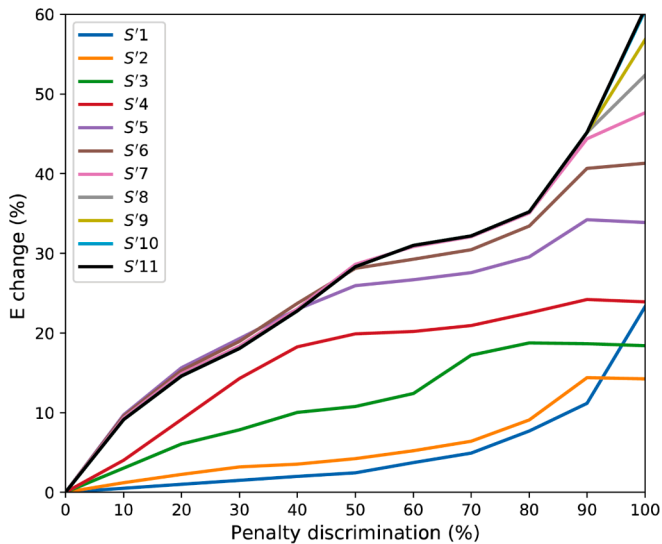


Fig. 7. Watershed-level total electricity supply (E) change (%) under varying penalty factor discrimination levels (p : %). Scenarios S'1 to S'11 represent the different levels of environmental constraint relaxation, with β_T increasing from 0°C to 2°C at the step of 0.2°C and β_w increasing from 0 to 14% at the step of 1.4%.

watershed agencies to decide which and how much constraints relaxation should be granted to each power plant and how much of a penalty should be imposed on constraints violation. Agencies' different preferences may lead to different policies and thus different environmental impacts and electricity supplies. For example, an aggressive agency that only values the reliability of the thermoelectricity supply system may allow plants to break the initial environmental constraints as much as needed to maintain the maximum electricity supply. An agency that chooses to reduce the cooling water consumption and thereby transfer more water supply to other sectors may set a small β_w and a large α_w . An agency that seeks to maintain a suitable water temperature for aquatic life may set a small β_T and a large α_T . The modeling of watershed-level decisions is warranted in the future.

4.4. Sensitivity analysis of electricity supply to the market electricity price

The market electricity price (EP) is a critical exogenous parameter in the water-electricity nexus model. In the above analysis, EP is assumed as a constant during the whole study period, which is referred to as the baseline scenario EP^0 . However, there are great daily variations in EP , ranging from 30 to 80 $\$/\text{MWh}$ during the 2012 summer (ICE, 2012). To analyze the sensitivity of electricity supply to EP , four scenarios are designed in this section. The daily EP s during the study period are obtained from the ICE database (ICE, 2012) and are linearly interpolated to fill data gaps, referred to as the first scenario EP_t^1 .

The electricity price in the baseline scenario is calculated as the average of the daily EP_t^1 :

$$EP^0 = \overline{EP_t^1} \quad (18)$$

In the second scenario EP_t^2 , the daily EP_t^1 are increased by 20%:

$$EP_t^2 = EP_t^1 * (1 + 20\%) \quad (19)$$

In the third scenario EP_t^3 , the daily EP_t^1 are decreased by 20%:

$$EP_t^3 = EP_t^1 * (1 - 20\%) \quad (20)$$

In the fourth scenario EP_t^4 , the daily EP_t^1 are increased by 20% if they are larger than the average price and decreased by 20% if smaller:

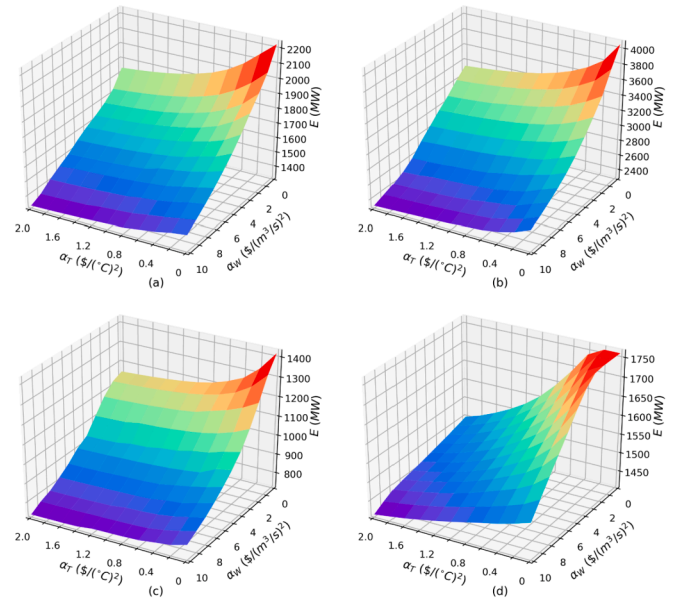


Fig. 8. Watershed-level electricity supply in different electricity price scenarios: (a) EP_t^1 , (b) EP_t^2 , (c) EP_t^3 , (d) EP_t^4 .

$$EP_t^4 = \begin{cases} EP_t^1 * (1 + 20\%), EP_t^1 \geq \overline{EP_t^1} \\ EP_t^1 * (1 - 20\%), EP_t^1 < \overline{EP_t^1} \end{cases} \quad (21)$$

The water-electricity nexus model is run over the four scenarios with all other parameters remaining the same as in Section 4.2. The watershed-level results are shown in Fig. 8. Comparing Fig. 8a to Fig. 4e, when the daily variations in the electricity price are considered, the potential electricity supply is reduced at most by 50%. This implicates more power generation curtailment during the days when the electricity price is lower. Compared to Fig. 8a, when the daily electricity prices are increased (decreased) by 20%, the potential electricity supply would be dramatically increased (decreased) in Fig. 8b (Fig. 8c). Comparing Fig. 8d to Fig. 8a, when there are larger daily variations in electricity price, the potential electricity supply is slightly decreased. The sensitivity analysis demonstrates that power plants are more willing to pay the economic penalties for more electricity generation when the *EP* is high and stable (Fig. 8a and 8b) but are less willing when the *EP* is low or unstable (Fig. 8c and 8d). The nexus modeling results prove to be very sensitive to *EP*. Therefore, much attention should be paid to determining a proper *EP* when the nexus model is used in future works. Modeling the electricity market equilibrium at the grid scale and obtaining an endogenous electricity price would advance the nexus model.

5. Conclusions and discussion

In this study, a water-electricity nexus model (PyHSPF-WE) is developed to support both water resources management and electricity supply planning during a drought event. The hydrologic module HSPF, the decentralized optimization module, and the thermoelectricity generation module are closely integrated at a daily time step at the watershed scale. The nexus model is capable of testing the various environmental regulations and economic penalties and quantifying the tradeoffs among streamflow, water temperature, and electricity supply at both the power plant and watershed levels. Findings show that some flexibility in environmental policies, i.e., relaxations of cooling water consumption and temperature constraints, would generally allow improvement in the reliability of the thermoelectricity supply system during a drought. Applying economic penalties can regulate the contributions of the individual power plants with different environmental impacts. The different responses of the power plants to various environmental policies, including environmental constraints relaxation and economic penalties on constraints violation, can be attributed to the different hydrologic conditions that exist at the plants' locations and their thermoelectric characteristics, such as primary fuels and cooling technologies. These general findings can be extended to other watersheds or drought events, even though the magnitude of the tradeoffs and the effects of environmental regulations and economic penalties may depend on the specific situation of the case study.

A case study is conducted in the Kaskaskia River watershed. The results show that the minimum required relaxations of cooling water consumption and temperature constraints are 14% and 3°C, respectively, if the power plants are expected to operate at their full capacities during the 2012 drought. At the watershed level, a spatially distinguished penalty policy, i.e., a larger penalty factor for upstream plants and a smaller one for downstream plants, is found to improve the total electricity supply by as much as 60% compared to the baseline penalty scenario. These results indicate that plant-specific policies have more advantages than uniform policies, although more efforts will be required for the watershed agencies to put these policies into practice.

The water-electricity nexus model developed in this study may be used for a wide range of applications, including planning and assessment and operational use. Our analyses in this study primarily focus on the planning and assessment of the tradeoffs between electricity supplies and environmental impacts during a drought event. For operational use with real-time hydrologic information unknown, weather forecast

information can be used as the climatic inputs to the HSPF module to predict the future streamflow and water temperature. One of the advantages of our nexus model is that it directly uses climatic inputs, and thus no streamflow inputs are required due to integration of the hydrologic model. We admit that the current nexus model would introduce uncertainties to hydrologic forecasts under an operational setting. How to incorporate the hydrologic uncertainties into water resources management is a typical issue investigated in many previous studies, e.g., Cai and Rosegrant (2004). Since the hydrologic uncertainties would increase with the lead time, in practice, we may suggest running the nexus model for a short period (e.g., a week rather than 3 months as in the current case study), and then running the model for the next period with an updated weather forecast. The data assimilation technique (Vrugt et al., 2005; Liu and Gupta, 2007) may be used to update the initial conditions of the next period with observed hydrologic data in the past period. The stochastic optimization technique (Salazar et al., 2013; Guerra et al., 2019) may also be used to consider the different hydrologic scenarios and provide the expected outcomes. The exact duration and extent of a drought event cannot be known before it ends under the operational setting. Based on the nexus model, however, environmental agencies can still be supported in obtaining useful hydrologic forecast information on the drought and setting predicted values for constraints relaxation and economic penalties.

A limitation of the proposed nexus model is that the electricity price is set as an exogenous variable, though a sensitivity analysis is conducted to demonstrate the responses of power plants to different price scenarios. We also assume that paying economic penalties for extra water consumption and a rise in the water temperature would not affect bidding offers of the power plants and the electricity price at the grid scale. This study focuses on the thermal power plants within a watershed with a relatively small number of plants and total generation capacity. From an economic point of view, in a market with many competitive sellers, the power sellers within such a watershed, which is much smaller than the region with a power grid, is usually not powerful enough to affect the market equilibrium price. If the model is applied to a much larger spatial scale, all these factors must be taken into account. Some future works may improve this newly developed water-electricity nexus model for more policy implications. For example, if the predictive daily or even hourly electricity demands and prices in the coming months are available, the dynamics of the electricity market and the response of each power plant can be represented with more details. This would require the incorporation of a system-level electricity market model that allows us to investigate the impacts of paying economic penalties on the bidding offers and finally the market electricity price by simulating the electricity demand, supply, and the equilibrium price in a consistent framework. The inclusion of a unit commitment and dispatch model (Fooladivanda and Taylor, 2015) has the potential to simulate the changes in electricity generation associated with electricity price and test the assumption of the exogenous price in the current case study. A more detailed thermoelectricity generation model at the power plant level would also help simulate the dynamic status of the generating units (Zhai et al., 2011). For example, dynamic cooling water consumption intensities and operating costs at each time step can be determined considering the ambient hydrologic conditions such as cooling water availability and river water temperature. The current nexus model focuses on the daily average analysis, mainly limited by the availability of hydrologic information. The nexus model can be enhanced to capture the sub-daily dynamics of the power generation and the associated environmental impacts, if sub-daily hydrologic conditions, e.g., hourly streamflow and river water temperature, can be simulated by climatic inputs with a finer temporal resolution or estimated by some assumptions on the sub-daily variations of streamflow.

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Appendix A. HSPF calibration and validation results

The HSPF model calibration and validation results are presented here. The Nash-Sutcliffe efficiency coefficient (NS: $-\infty \sim 1$) (Nash and Sutcliffe, 1970) is used to evaluate the model performance. A higher NS means a better model performance.

Streamflow is calibrated during 1980–1995 and validated during 2001–2012 at the gage station USGS 05594100. Fig. A1 shows the results of simulated daily streamflow for calibration (Fig. A1a) and validation (Fig. A1b). The NS coefficients are 0.721 and 0.635, respectively, which are adequate for capturing the streamflow variations in the watershed. However, a good calibration of the model at that station does not necessarily guarantee the model accuracy in other sections of the basin.

Fig. A2 shows the results of simulated daily water temperatures. Observed data for water temperatures (T) are not as adequate as those for streamflow. Only a small amount of observation records are available for the Kaskaskia River watershed (USGS, 2018), especially after the year 2000. For this reason, we calibrate (Fig. A2a) the HSPF water temperature module with observed data from the gage station USGS 05594100 during 1980–1995, while validating (Fig. A2b) for the same period with observed data from the 15 other USGS stations. This validation strategy is like one-round cross-validation. HSPF shows a good performance for simulating daily water temperature, with NS as 0.917 and 0.882 for calibration and validation, respectively.

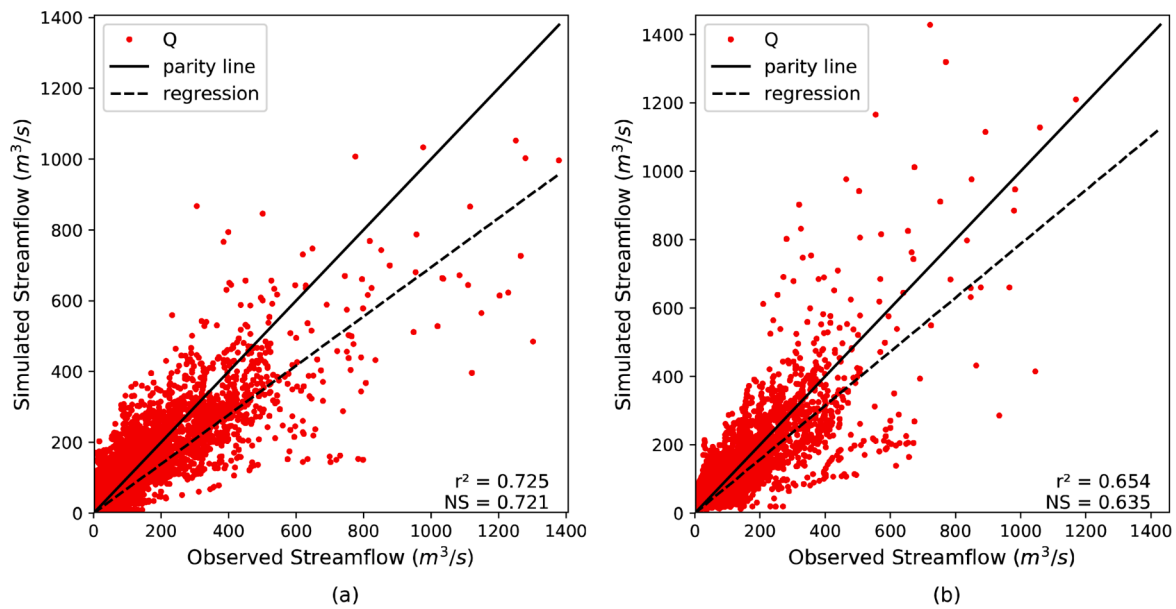


Fig. A1. Evaluation of HSPF simulated daily streamflow ($Q: m^3/s$): a) calibration and b) validation at the gage station USGS 05594100.

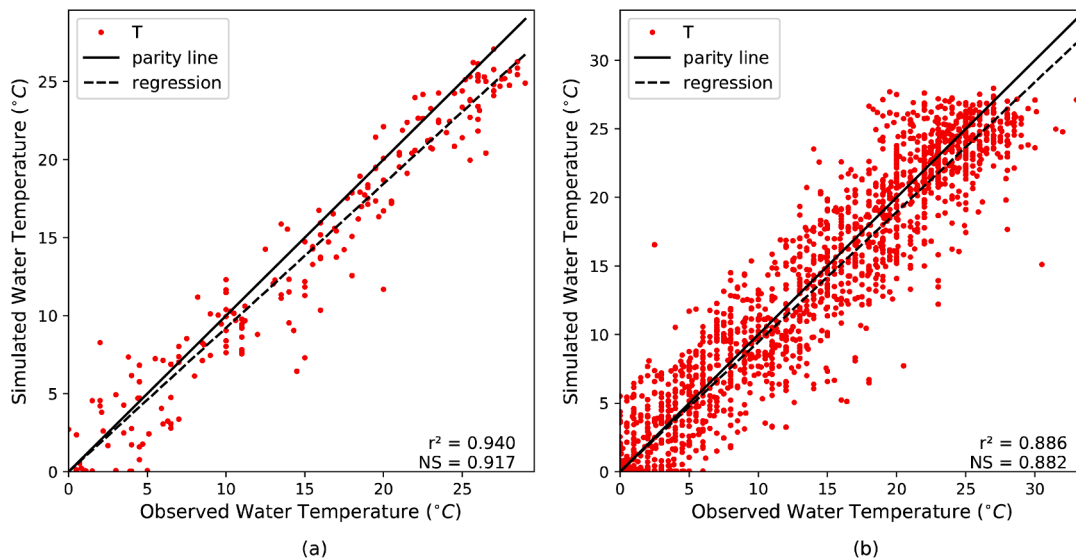


Fig. A2. Evaluation of HSPF simulated daily river water temperature ($T: ^\circ C$): a) calibration at the gage station USGS 05594100, and b) validation at the other 15 USGS stations, both during 1980–1995.

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